



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

– Hira Kumar

ISP (Internet service provider)



- ISP stand Internet Service Provider.
- An ISP is a company or organization that provides individuals, businesses, and institutions access to the internet.
- An ISP can provide internet services the gateway/bridge between your local device (computer, smartphone or router) and the global internet network through wired or wireless technologies.
- ISPs also provide services such as IP addresses, DNS, and data transmission.
- ISPs are also responsible for providing a public IP to your device, so that it can access the Internet.
- Most of the ISPs bundle their internet services with other services like telecommunications, OTT streaming, or Dish TV to attract more customers.
- Examples of ISPs in India include **Airtel, Jio, BSNL, and Vi.**

❖ History of ISP

- The first commercial ISP, The World, launched in 1989 in the U.S., offering dial-up internet access. By the mid-1990s, ISPs like AOL and EarthLink popularized home internet.
- Broadband replaced dial-up in the 2000s, with speeds jumping from 56 Kbps to over 100 Mbps. Today, fiber-optic ISPs like Google Fiber and JioFiber offer gigabit speeds.

❖ Role of ISP

1. **Provides Internet Connectivity** – Connects users to the Internet through fiber, cable, wireless, DSL, etc.
2. **Assigns IP Addresses** – Provides IP addresses to identify devices on the network.
3. **Provides DNS Services** – Translates domain names into IP addresses.
4. **Manages Network Traffic** – Routes and manages data between users and Internet networks.
5. **Provides Additional Services** – May provide email, web hosting, VPN, cloud, and other services.

❖ How Does an ISP Provide Internet Access

Here is a simple technical step-by-step process that shows how you access the internet through an ISP.

Step 1: Your Device Connects to the ISP network

Your device (phone, laptop, etc.) connects to your ISP's network via a medium like WiFi, mobile data, or Ethernet.

Step 2: ISP assigns an IP Address to your device

The ISP assigns your device a unique IP address, which identifies it on the internet and enables communication with other devices and servers.

Step 3: DNS Resolution

When you enter a website (e.g., www.google.com), your ISP uses the Domain Name System (DNS) to translate the domain name into a numerical IP address. However, different ISPs may depend on different DNS servers.

Step 4: Routing Data Between Your Device and the Internet

Using protocols like BGP (Border Gateway Protocol), your ISP routes your data through its network and across the internet to the destination server.

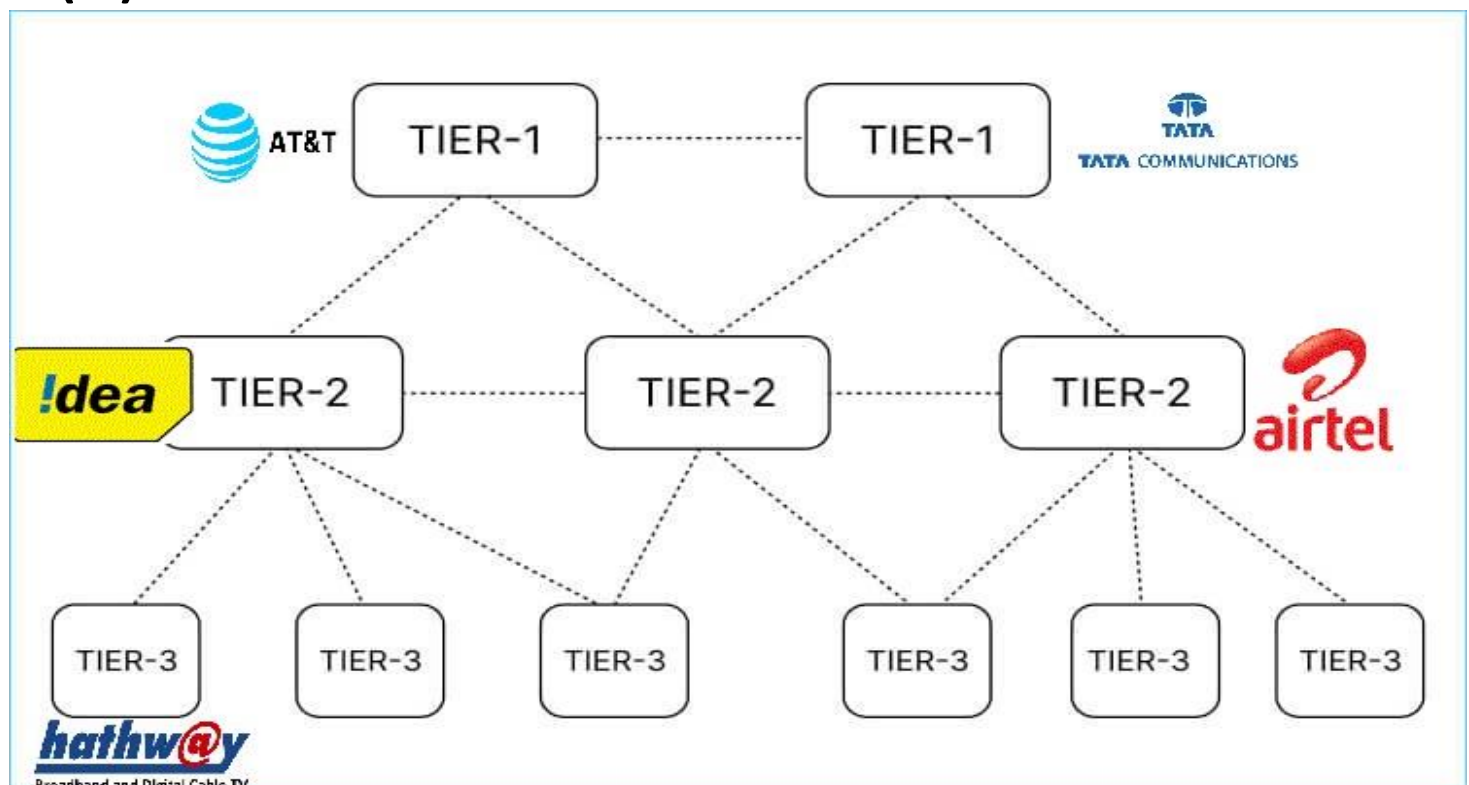
Step 5: Content Delivery

The destination server sends the requested data (e.g., a webpage or video) back through the same route, reaching your device via the ISP's network.

❖ Types of ISP based on Tier system

Total 3 types of tier system

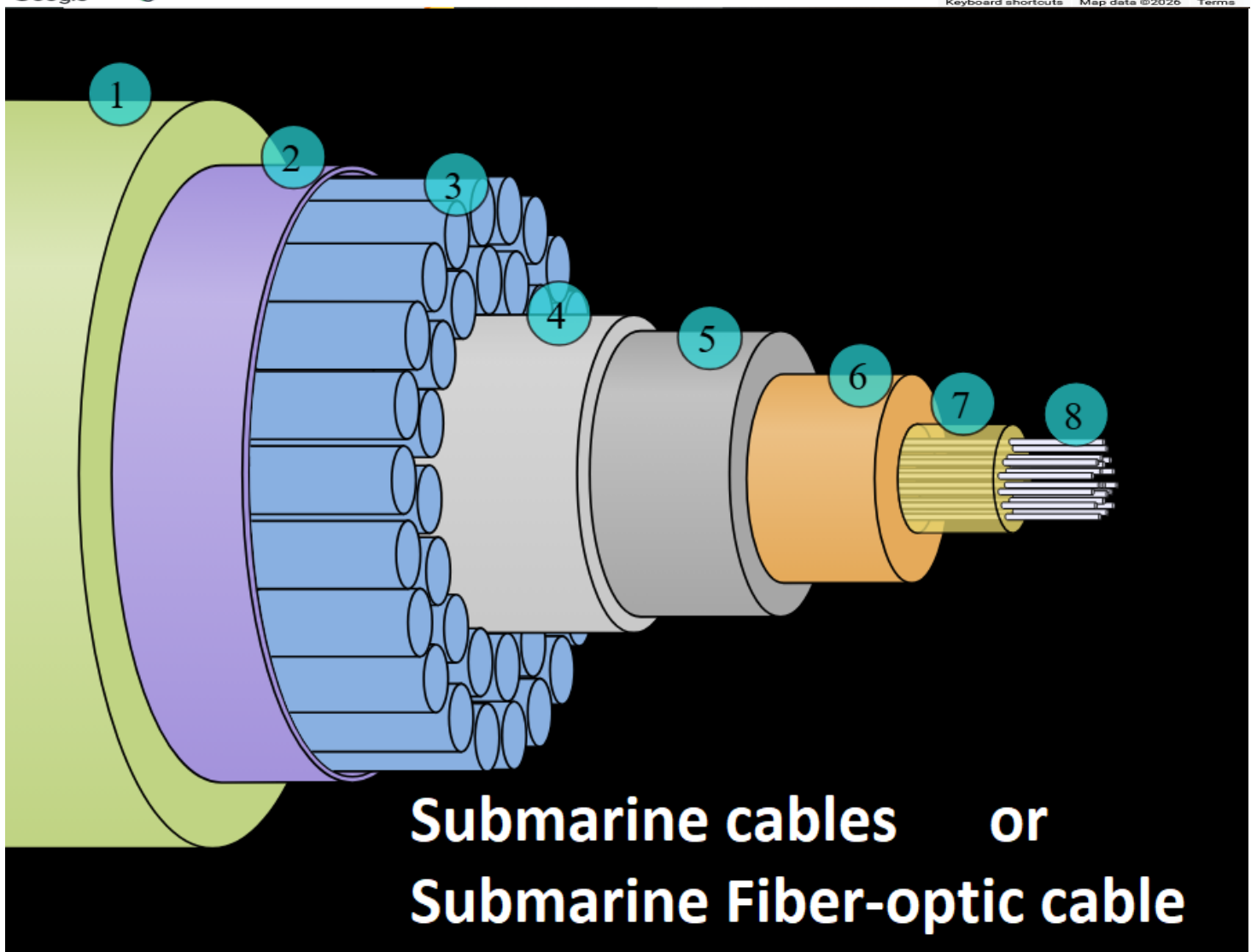
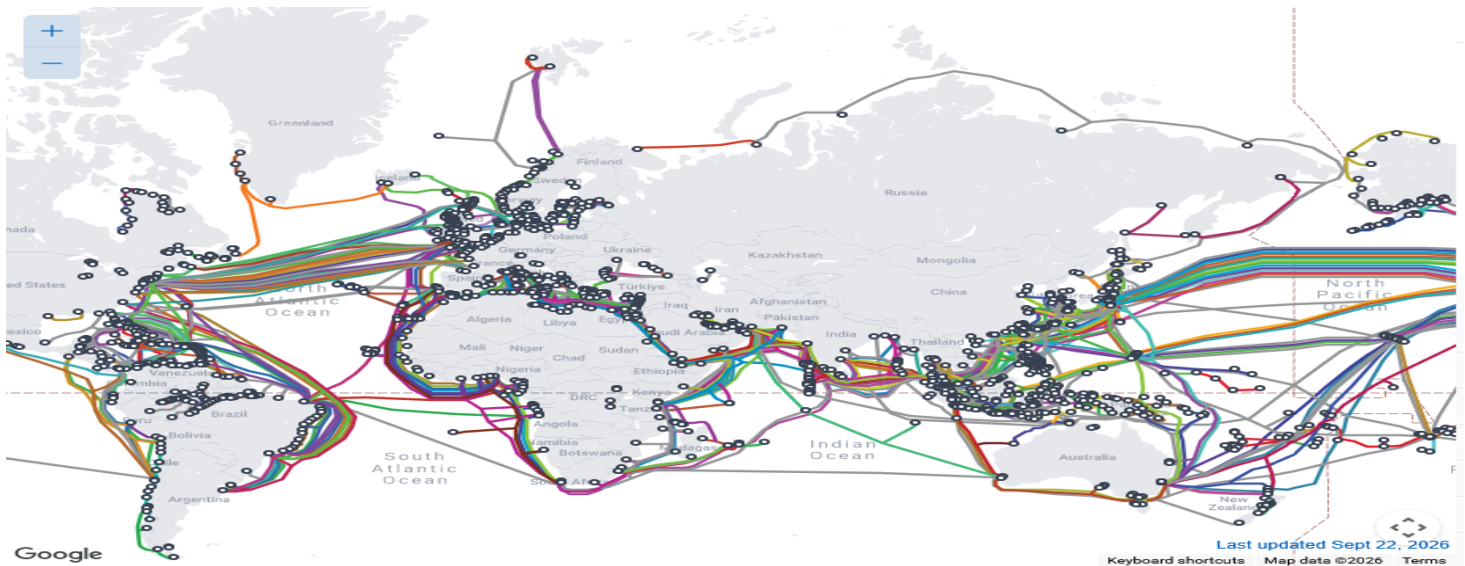
- (i) Tier-1
- (ii) Tier-2
- (iii) Tier-3



(i) Tier 1 ISPs (The internet backbone)

- Tier 1 providers form the core infrastructure of the global internet. They own massive physical assets, including transoceanic submarine cables / undersea fiber-optic cable and high speed terrestrial fiber routes.

- Example : AT&T, Tata communications, Lumen, cogent communications, NTT,...





The following companies in India are owners or operators of international submarine fiber-optic cables.

Company	Cable	Main route / connects
Reliance Jio	IAX (India-Asia-Xpress)	India → Singapore → Southeast Asia
Reliance Jio	IEX (India-Europe-Xpress)	India → Middle East → Europe
Bharti Airtel	i2i	India (Chennai) → Singapore
Bharti Airtel	SEA-ME-WE 4	Southeast Asia → Middle East → Mediterranean/Europe
Bharti Airtel	SEA-ME-WE 6	Singapore/Asia → Middle East → Europe
Bharti Airtel	2Africa	Africa ↔ Middle East ↔ India/Asia ↔ Europe
Tata Communications	TGN-Tata Indicom (TIC)	India (Chennai) → Singapore

Company	Cable	Main route / connects
Tata Communications	SEA-ME-WE 3/4, SAFE, FLAG, etc.	India connected with Asia, Middle East, Africa and Europe
GCX	FALCON	India → Middle East → Europe
GCX	WARF	India → Sri Lanka → Maldives
Sify	MENA / GBI	India → Middle East / Europe-Africa connectivity

See location & cables : <https://www.submarinecablemap.com/>

(ii) Tier-2

- Tier 2 providers operate on national or multi-region scales. While they build extensive regional infrastructure, they lack complete global reach on their own.
- They act as the middle layer, connecting local last -mile networks to global backbones and large enterprise clients.
- Example : Reliance Jio, Bharti airtel, Vodafone, Comcast, british telecom,...

Top Internet Service Providers Across the World

#	Country	Leading ISP	#	Country	Leading ISP
1.	USA	Verizon	25.	Denmark	TDC Net
2.	India	Reliance Jio	26.	Chile	Entel
3.	China	China Mobile	27.	Belgium	Proximus
4.	Indonesia	Telkomsel	28.	Israel	Partner Communications
5.	Turkey	Turkcell	29.	Kenya	Safaricom
6.	Japan	NTT Docomo	30.	Saudi Arabia	STC
7.	France	Orange	31.	Portugal	MEO
8.	Germany	Deutsche Telekom	32.	Croatia	Hrvatski Telekom
9.	United Kingdom	EE	33.	Sweden	Telia
10.	Australia	Telstra			
11.	Mexico	Telcel			
12.	Spain	Movistar			
13.	Netherlands	KPN			
14.	Thailand	AIS			
15.	Malaysia	Maxis			
16.	Singapore	Singtel			
17.	Greece	Cosmote			
18.	Vietnam	Viettel			
19.	Hong Kong	HKT			
20.	Bangladesh	Grameenphone			
21.	Norway	Telenor			
22.	Finland	Elisa			
23.	Poland	Orange Polska			
24.	Switzerland	Swisscom			



The following companies in India are owners or operators of fiber-optic cables inside india.

Company	Major Fiber Network Presence
GTPL Hathway	Gujarat, West Bengal, Rajasthan, Maharashtra and other states
Hathway Cable & Datacom	Maharashtra, Gujarat, Delhi-NCR, Karnataka, Telangana, Andhra Pradesh, Tamil Nadu, West Bengal, Madhya Pradesh and others
ACT Fibernet	Karnataka, Telangana, Tamil Nadu, Delhi-NCR and other major cities
Excitel	Delhi-NCR, Telangana, Karnataka, Maharashtra, Uttar Pradesh, Rajasthan and other cities
Netplus Broadband	Punjab, Haryana, Himachal Pradesh, Rajasthan and other North Indian regions
Alliance Broadband	West Bengal and several other states
Spectra	Delhi-NCR and selected other cities
Kerala Vision Broadband	Kerala
AP FiberNet	Andhra Pradesh
RailTel	Pan-India optical-fiber network





RAILTEL

Railtel Corporation Of India Limited

Company	Presence in Bihar
BSNL	Bihar — large fixed-line/FTTH network
RailTel	Bihar and pan-India backbone
Airtel	Bihar, including Patna and other cities
Reliance Jio	Bihar, including Patna and other cities
GTPL	Patna and other areas
Local/Regional ISPs	Various cities and districts

(iii) Tier-3

- A Tier-3 ISP is a small or local Internet Service Provider that mainly provides Internet connectivity to end users in a limited geographic area, such as a city, district, or group of villages. It usually purchases Internet transit or connectivity from Tier-1 or Tier-2 ISPs rather than connecting directly to the global Internet backbone.
- Example :
 - ✓ **Netplus Broadband** — Punjab & nearby regions
 - ✓ **Wishnet Broadband** — Kolkata, West Bengal
 - ✓ **FusionNet** — selected cities/regions
 - ✓ **Local city-level fiber ISPs** — Bihar, Uttar Pradesh, Rajasthan, etc.
 - ✓ **Local cable/fiber broadband operators** serving a particular town or

❖ Types of ISP Services

Based on your requirements or location, an ISP can offer different types of internet services:

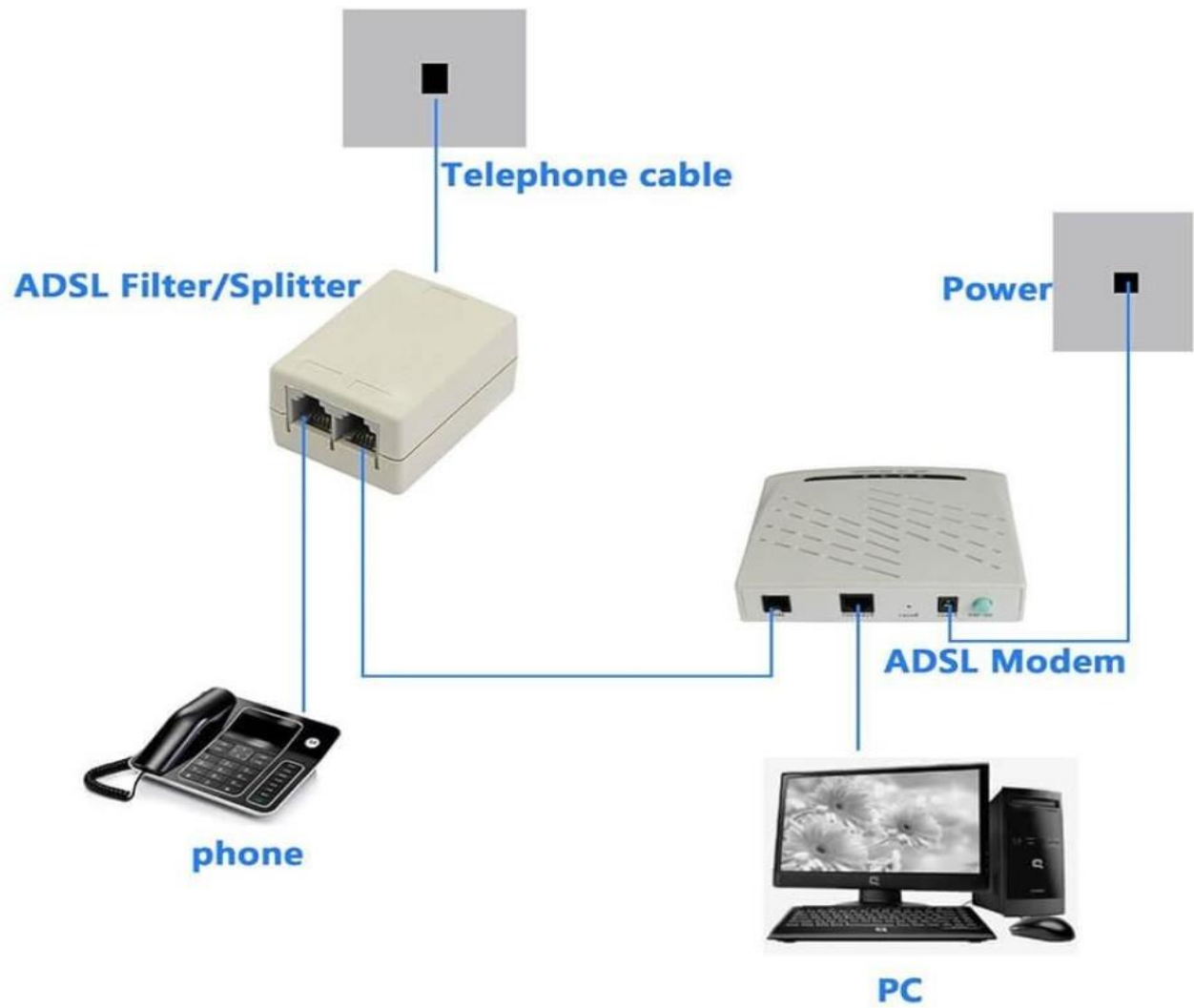
ISP Service	Description
Broadband	High-speed internet, faster than dial-up.
DSL	Uses phone lines, moderate speeds.
Fiber-optic	Very high speeds via light signals.
Cable	Faster than DSL, slower than fiber.
Satellite	Ideal for rural areas, slower with higher latency.

(i) ISP Broadband Service

- An ISP broadband service provides high-speed Internet access to customers through technologies such as fiber-optic cables, coaxial cables, DSL, or wireless networks. The ISP connects customers to the wider Internet and manages bandwidth, IP addresses, authentication, billing, and network maintenance to provide reliable Internet connectivity for homes and businesses.

❖ What is DSL

- **DSL (Digital Subscriber Line)** is a broadband technology that provides Internet access through existing copper telephone lines. It allows users to make telephone calls and use the Internet simultaneously by using different frequency ranges.
- Flow : ISP → Telephone Exchange → Copper Telephone Line → Customer Modem/Router



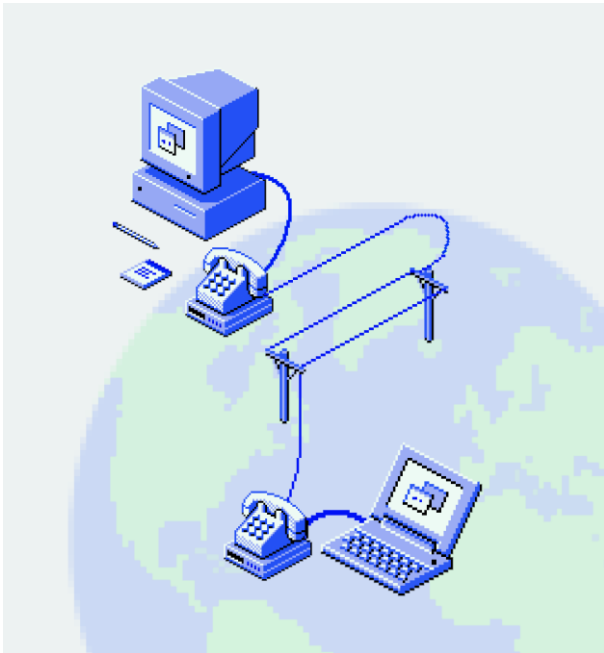
❖ What is Dial-up connection

- A dial-up connection is an older method of accessing the Internet using

HIRA KUMAR

a telephone line and a modem. The modem dials the ISP's telephone number to establish a connection. It provides very low Internet speeds, and the telephone line generally cannot be used for normal voice calls while connected.

- Flow : Computer → Dial-up Modem → Telephone Line → ISP → Internet



❖ Difference Between Dial-up and DSL

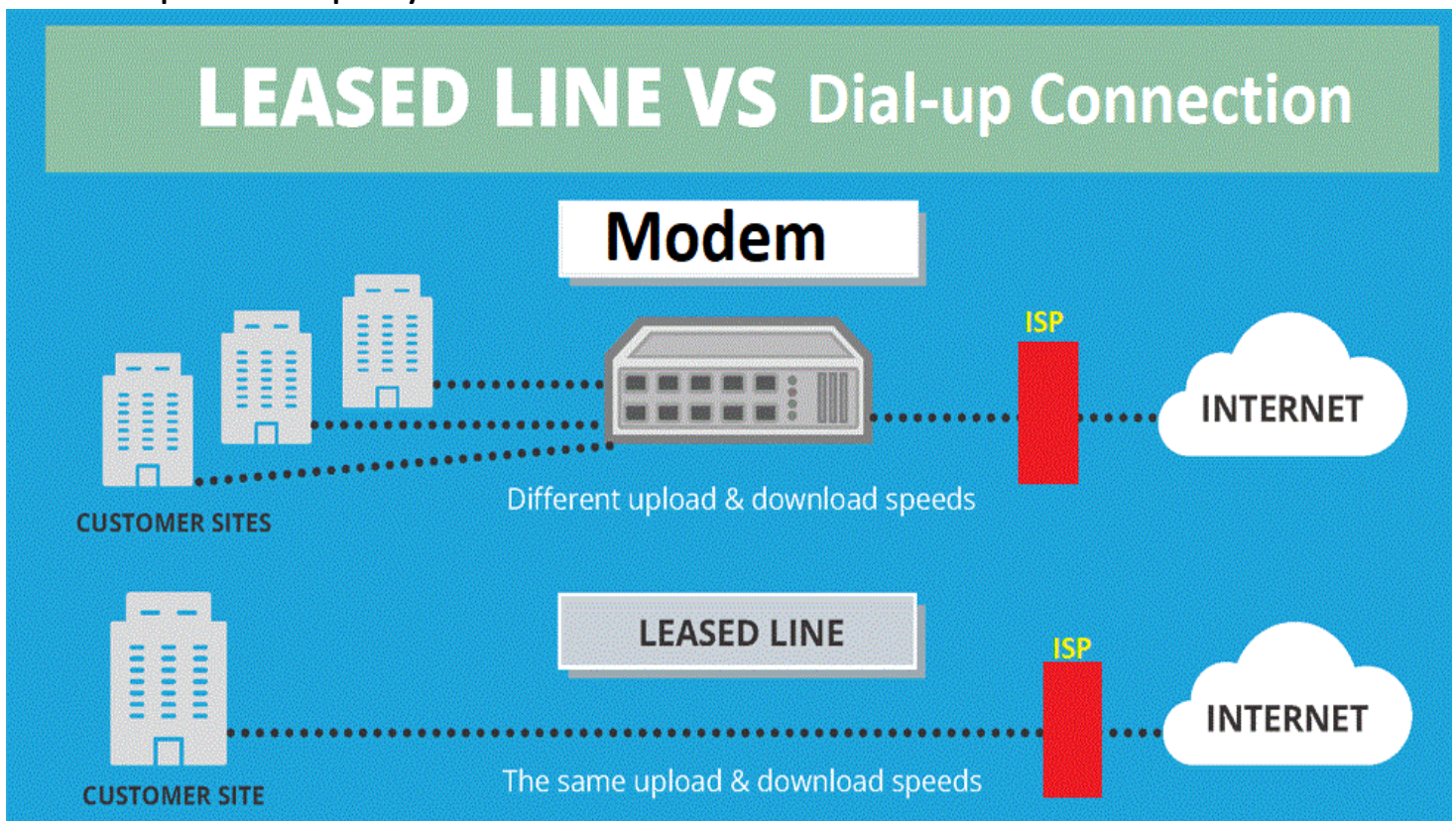
Feature	Dial-up	DSL
Full Form	Dial-up Internet Connection	Digital Subscriber Line
Medium	Telephone line	Telephone copper line
Speed	Very slow	Much faster
Connection	Modem dials the ISP's number	Always-on connection
Phone Calls	Usually cannot use Internet and phone simultaneously	Can use Internet and phone simultaneously
Technology	Uses the telephone network's voice channel	Uses higher-frequency signals on the telephone line
Current Use	Mostly obsolete	Also declining, but still used in some areas
Example	56 Kbps dial-up	ADSL, VDSL

❖ What is a Leased Line?

- A leased line is a dedicated, private Internet or communication

connection provided by an ISP between two locations, such as a company office and a data center. It provides guaranteed bandwidth, high reliability, low latency, and a fixed connection.

- Monthly price approx : 50,000rupees /month
- Example: Company Office → ISP Network → Data Center



❖ Fiber connection

- A fiber-optic connection provides Internet access through thin optical fibers. Data is transmitted as light signals, allowing very high speed, long-distance communication, and low latency.
- Flow: ISP → Fiber Network → Optical Fiber Cable → ONU/ONT → Wi-Fi Router → User Device
- Example: JioFiber, Airtel Xstream Fiber, BSNL FTTH.

❖ Cable Connection

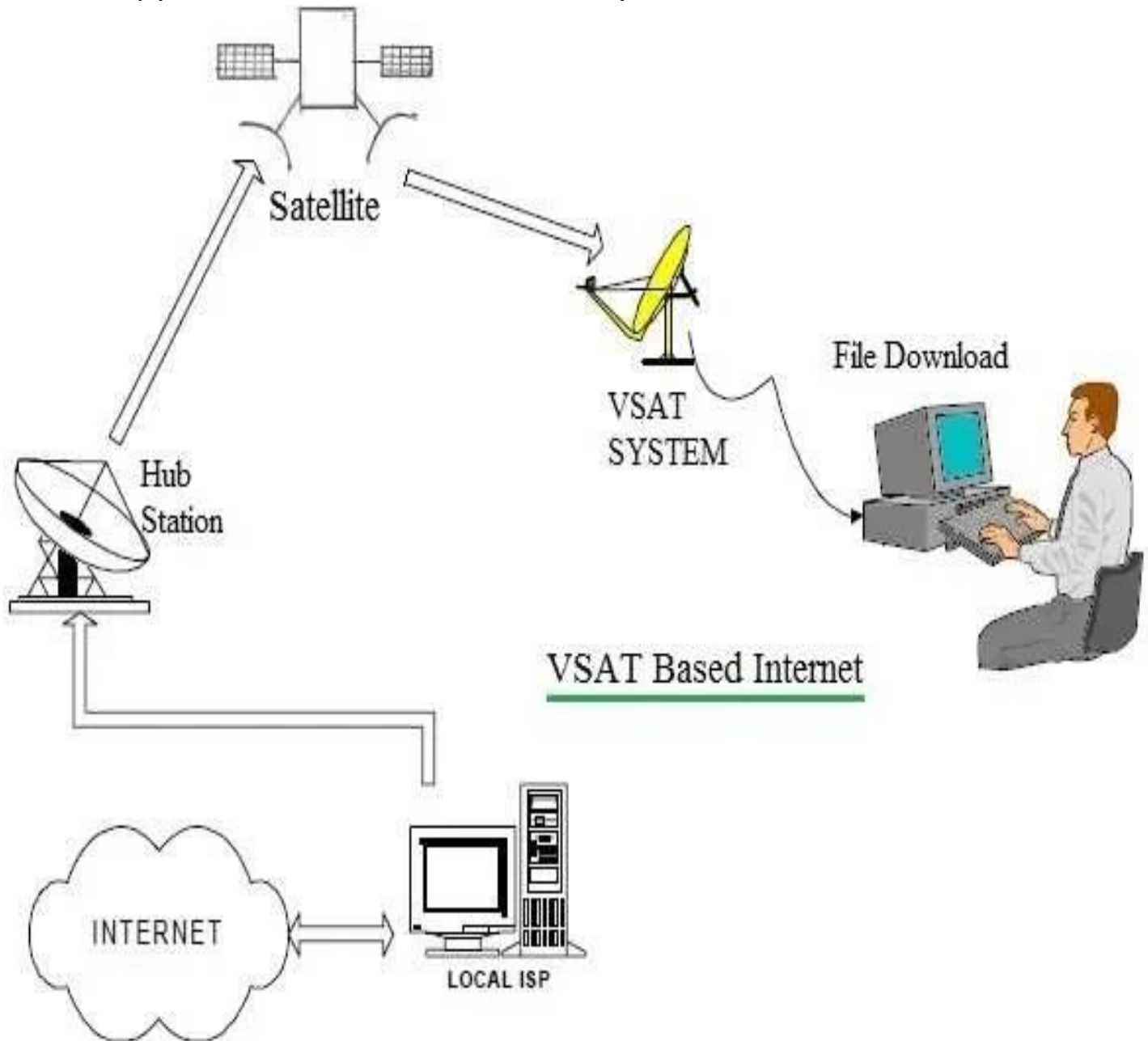
- A cable Internet connection provides Internet access through coaxial cables or hybrid fiber-coaxial (HFC) networks. Data is transmitted using electrical/radio-frequency signals.
- Flow: ISP → Fiber/Headend → Coaxial Cable → Cable Modem → Wi-Fi Router → User Device

(ii) What is VSAT?

- VSAT (Very Small Aperture Terminal) is a satellite-based

communication system that uses a small dish antenna to send and receive data through a satellite.

- It is commonly used to provide Internet, voice, and data connectivity in remote areas where fiber or terrestrial networks are unavailable.
- **Simple flow:** User/Office → VSAT Dish → Satellite → Ground Station/Internet
- **Example:** Remote villages, ships, oil fields, ATMs, and disaster-affected areas, army camp can use VSAT connectivity.
- Price : approx. 25,000/ month for 5mbps

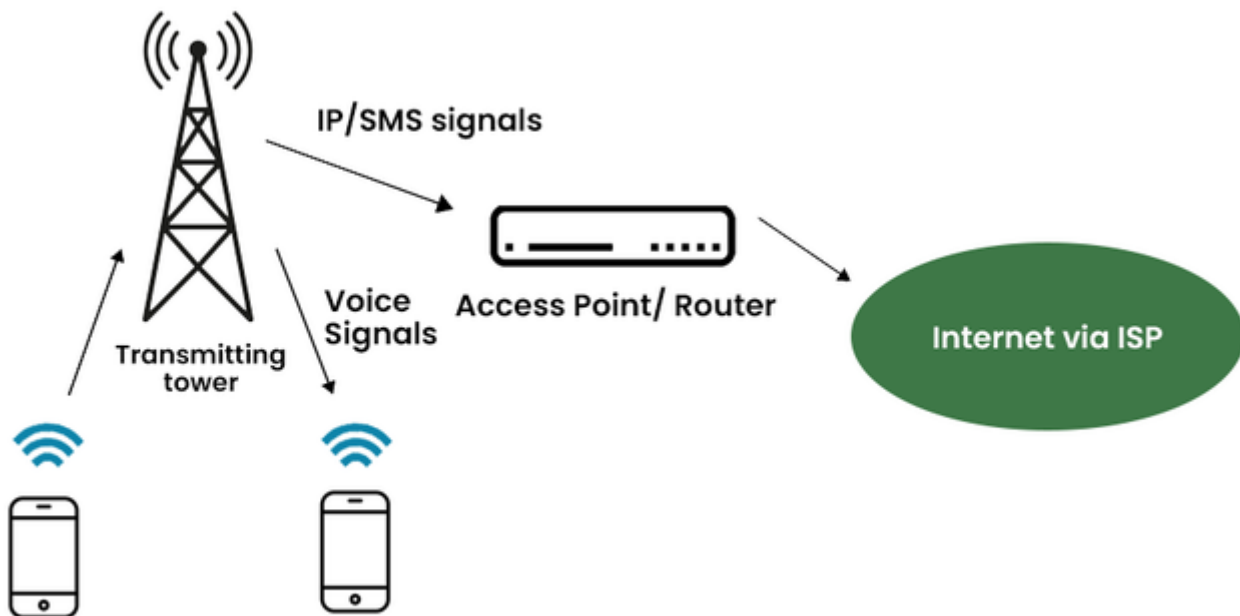


(iii) Mobile Internet

- Mobile Internet is also provided by an ISP/telecom operator, but the connection reaches your phone through a cellular network rather than

a fiber cable directly to your phone.

- How it works : Mobile Phone → Cell Tower → Mobile Operator's Core Network → Internet → Website/Server
- For example, Jio, Airtel, and Vi provide mobile Internet services.
- The cell tower connects your phone wirelessly, while the operator's backhaul/core network carries the data toward the Internet. Fiber is often used between the tower and the operator's network.
- Technologies: 4G LTE, 5G, etc.



❖ Other service through ISP

- (i) **IP address** : An ISP assigns a unique public or private IP address to identify and connect your device to the Internet.
- (ii) **DNS Services** : An ISP provides DNS servers that translate domain names, such as google.com, into their corresponding IP addresses.
- (iii) **Email services** : An ISP may provide email accounts and services for sending, receiving, and storing electronic messages.
- (iv) **Web hosting** : An ISP may provide server space and resources to host websites and make them accessible on the Internet.
- (v) **VPN/Enterprise services** : ISPs provide secure VPN and dedicated network services to connect offices, branches, employees, and business networks.
- (vi) **Cloud Services** – Provides cloud storage, computing resources, and online applications to customers.

- (vii) **Data Center Services** – Provides secure facilities for hosting servers, applications, and business data.
- (viii) **Backup Services** – Provides online backup and storage for protecting important customer data.
- (ix) **IoT Connectivity** – Provides Internet connectivity for smart devices, sensors, and connected machines.
- (x) **Cybersecurity Services** – Provides firewall, DDoS protection, antivirus, and other network security services.
- (xi) **VoIP Services** – Enables voice calls over the Internet instead of traditional telephone networks.

❖ **Advantages of ISP**

- They provide consistent and stable internet connections, essential for both personal and business use.
- ISPs offer different plans with varying speeds, allowing users to choose a package that fits their browsing, streaming, or business needs.
- Most ISPs offer customer service and technical support to quickly resolve internet issues or equipment problems.
- Many ISPs bundle internet access with other services like email hosting, cloud storage, VoIP (Voice over IP), and television packages.

❖ **Disadvantages of ISP**

The following are the disadvantages of an ISP:

- Due to intense market competition, some ISPs have been reported to engage in questionable practices like hidden charges, unexpected price hikes, and inconsistent service quality.
- Using an ISP makes you fully reliant on them for internet access, speed, security, and support, leaving little room for flexibility or control.
- If your ISP hosts too many users on the same shared server, it can lead to slower internet speeds, reduced website performance, and inconsistent connectivity.

What is ISP ? Discuss the role of ISP.

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=====HIRA KUMAR=====